

Demonstrating JumpMod: Haptic Backpack that Modifies Users' Perceived Jump

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ABSTRACT

Vertical force-feedback is extremely rare in mainstream interactive experiences. This happens because existing haptic devices capable of sufficiently strong forces that would modify a user's jump require grounding (e.g., motion platforms or pulleys) or cumbersome actuators (e.g., large propellers attached or held by the user). To enable interactive experiences to feature jump-based haptics without sacrificing wearability, we propose JumpMod, an untethered backpack that modifies one's sense of jumping. JumpMod achieves this by moving a weight up/down along the user's back, which modifies perceived jump momentum—creating accelerated & decelerated jump sensations. Our device can render five distinct effects: jump higher, land harder/softer, pulled higher/lower, which we demonstrate at SIGGRAPH 2023 Emerging Technologies in two jump-based VR experiences.

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1 INTRODUCTION

Enabling interactive devices to move the user's entire body (i.e., body-scale force-feedback) has proven challenging for researchers and practitioners (e.g., theme-parks or VR arcade designers). The typical approach is to install large actuators in the environment, such as motion platforms [Rheiner, 2014], mechanically-actuated pulleys [Eidenberger and Mossel, 2015], or attach powerful propellers to the user's body [Sasaki et al., 2019; Takahashi et al., 2018]. While these are effective since their output force is large, they limit entirely users' mobility—these devices are at odds with users' home setups (e.g., free-walking VR experiences [Matsumoto et al. 2016])

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Figure 1: (a) We propose JumpMod, a wearable device that modifies the user's perceived jump. It achieves this by moving a weight along the user's back, causing an inertial force that modifies one's sense of vertical momentum. JumpMod renders five distinct sensations: a sense of (b) jumping higher; (c) being pulled lower; (d) landing harder or (e) softer; and (f) being pulled higher

and are at odds with experiences where users are highly mobile (e.g., interactive sports [Kunze et al., 2017]). Instead, to enable more mobile contexts, a vertical body-scale actuator that is untethered and wearable would prove ideal.

To tackle this and expand the portability of vertical force-feedback, we propose JumpMod, an untethered backpack that modifies the user's perceived jump. Our device, depicted in Figure 1, achieves vertical effects by moving a weight along the user's back at different phases of a canonical jump (launching, ascending, descending, landing). The key behind JumpMod is that it does not physically lift the user as expected by using motors and infrastructure 10x larger/heavier. Instead, JumpMod creates only a sufficiently strong inertial force that users can feel, which modifies their perceived jump, all while keeping our device relatively small.

2 IMPLEMENTATION

JumpMod is a self-contained haptic backpack (i.e., works on battery and is wireless), as shown in Figure 2. The battery serves a dual purpose, both as a power source and as a moving weight. Jumping effects are rendered by moving the weight at high speeds along a linear rail using a brushless DC motor and a timing belt. Constrained by the rail, the weight can only move in one dimension, and its position is continuously tracked using a rotary magnetic encoder. Two limit switches are located at the ends of the rail—if they are

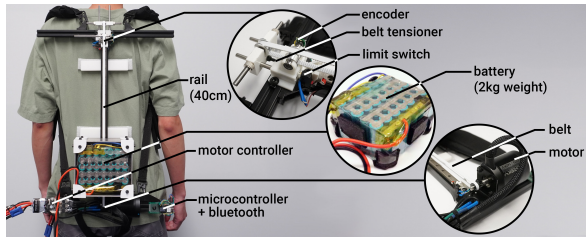


Figure 2: Hardware implementation of our untethered and wearable haptic device.

triggered or if the encoder detects a fault, the microcontroller stops the motor to prevent any potential hazards. Since microcontrollers can also be affected by faults, metallic physical stoppers added at the rail's ends ensures that the weight stays inside the rail even if the limit switches do not trigger. More details on the implementation, characteristics and evaluation of our device can be found in [Nith et al., 2023].

3 USER EXPERIENCE

To illustrate the applicability of our device at SIGGRAPH 2023 Emerging Technologies, we demonstrate two jumping-based experiences in VR wearing our untethered backpack and a wireless VR system: (1) an endless runner game and (2) an escape room game.

3.1 VR endless runner

The VR endless runner experience is set in a farm scene shown in Figure 1(a), and the game is played by running and jumping in place to avoid obstacles and collect points by smashing pumpkins. Our device enhances the experience by rendering different types of jumping effects:

- **Jumping higher.** By collecting a virtual jump boost power-up, when the user jumps, our device actuates its weight *up* in sync with the user's *leaving the ground*, as depicted in Figure 1(b). This extra inertial force from the moving weight creates a sense of *jumping higher*. In addition to *seeing* their avatar jumping over the cow, they also *feel* this matches the *physical sensation* of jumping higher.
- **Pulling down.** Figure 1(c) illustrates the user jumping over a haystack. However, when the user is about to land, the wind blows harder and pulls the user down. This "pulling" sensation is rendered by moving the device's weight *down* as the user *lands on the ground*.
- **Landing softer.** When the user jumps in a mud puddle, our device renders this "softer" landing compared to solid ground by moving its weight *up* when the user *lands*.
- **Landing harder.** When the user jumps on top of a pumpkin, as shown in Figure 1(d), our device moves the weight *down* when the user *descends* (past the apex, the highest point of their jump), rendering the sensation of smashing the pumpkin.
- **Pulled higher.** Figure 1(f) depicts the user getting help from a bird as it flies next to the user and pulls them up for a higher jump. Our device renders this "pulled higher" by moving the weight *down* at the user's *launch*.

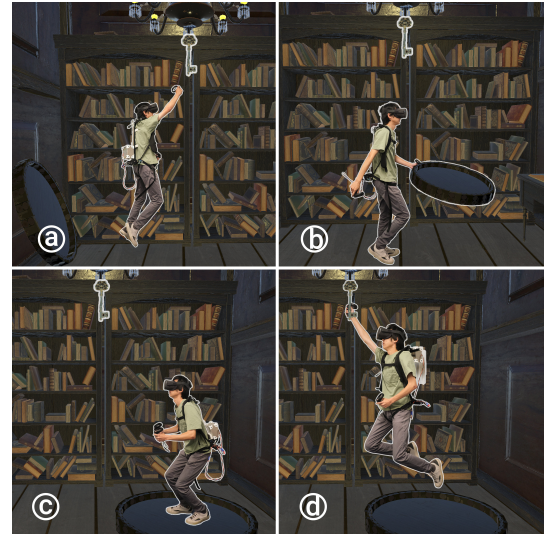


Figure 3: (a) The key is out of reach even if the user jumps to their fullest. (b) They find and move a trampoline, so that (c) by jumping on it (d), they can jump higher and grab the key—they *see* themselves jumping higher and also *feel* it with our device.

3.2 Free-walking & controller-based VR escape-room

Our second application demonstrates a VR *escape room*; the audience will solve puzzles by walking and jumping. Figure 3 depicts the interaction in the first of four rooms we designed: (a) the key to solving the puzzle is out of reach even if the user jumps; (b) they find a trampoline, which they grab using the VR handheld controllers; (c) by jumping off the trampoline, (d) they jump higher and grab the key—not only they see a higher jump, but our device creates the feeling of jumping higher. Note that our device is able to render these vertical sensations without encumbering the user's walking.

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